

Final Machine Learning Project Analysis Report

Project Title: Stroke Prediction Classification Analysis

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Course: ITAI1371 - Machine Learning

1. End-to-End Approach

The objective of this project was to build and evaluate a robust machine learning classification pipeline to predict whether a patient will experience a stroke.

Data Preprocessing & Splitting:

The dataset was loaded, dropping missing values and unnecessary identifiers. To ensure unbiased evaluation, the data was strictly split into Train (70%), Validation (15%), and Test (15%) sets using stratified sampling to preserve the target class distribution.

Feature Engineering & Transformation:

Numerical features (age, hypertension, heart_disease, avg_glucose_level, bmi) were imputed using the median and scaled using StandardScaler. Categorical features (gender, ever_married, work_type, Residence_type, smoking_status) were handled with missing-value imputation and one-hot encoding via scikit-learn's ColumnTransformer.

Model Training:

Six distinct classification models were implemented and trained:

Logistic Regression

Decision Tree Classifier

Random Forest Classifier

Gradient Boosting Classifier

K-Nearest Neighbors (KNN) Classifier

Support Vector Classifier (SVC)

Ensemble Modeling:

Following individual evaluations, an Averaged Probability Ensemble and a Bayesian Ensemble were built using the top three performing models to improve generalization and prediction confidence.

2. Handling Class Imbalance with SMOTE (Training Set Only)

An exploratory analysis of the stroke dataset revealed severe class imbalance: approximately 95% of the records represented non-stroke cases (0), while only 5% represented actual stroke cases (1).

Why SMOTE was Required:

Without intervention, standard classifiers achieve a high apparent accuracy (~95%) simply by predicting the majority class ("no stroke"), resulting in a near-zero recall and F1-score for the minority class. To resolve this, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to generate synthetic examples of the minority class.

The "Training Set Only" Rule:

Crucially, SMOTE was integrated using an imbalanced-learn pipeline (ImbPipeline) so that over-sampling occurred strictly within the training folds during `.fit()`. The Validation and Test sets were left entirely untouched. If SMOTE were applied to the entire dataset before splitting, synthetic data points generated from validation/test distributions would leak into the training phase, artificially inflating performance metrics and corrupting the evaluation framework.

3. Best-Performing Model Analysis

Based on the comparative evaluation across validation and test sets, the Random Forest Classifier and the Bayesian Ensemble achieved the top performance scores (Test ROC-AUC: 0.9100, Test F1-Score: 0.3500).

Why Random Forest Excelled:

Random Forest handles non-linear relationships and complex interactions between clinical features (such as age, average glucose levels, and hypertension) exceptionally well through an ensemble of randomized decision trees.

Impact of Ensembling:

The Bayesian Ensemble matched the top individual model by dynamically weighting model probabilities based on validation ROC-AUC scores, successfully dampening individual model variance and producing stable probability outputs.